

VIKHYAT CHAUHAN

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SUMMARY

I build agentic AI systems and the tooling around them — LangGraph agent pipelines, two-stage RAG, locally-run LLMs — and I ship end to end. Founder turned production engineer: I bootstrapped a hardware startup solo (PCB to cloud to app, 100+ homes), then optimized TensorRT inference on FDA-regulated systems at GE. I move fast, own the whole stack, and reason carefully about where the human stays in the agent loop.

SKILLS

- **AI & Agentic Systems:** agentic workflows (LangGraph, LangChain), tool use/function calling, RAG (two-stage retrieval, BGE embeddings, cross-encoder reranking), LLM fine-tuning (LoRA), LLM-as-judge / evals, semantic caching, MCP, spec-first agent harnesses / Claude Code workflows, OpenAI APIs
- **ML & Inference:** PyTorch, TensorFlow, TensorRT (ONNX export, FP16/INT8 precision tuning, custom CUDA/C++ plugins, layer fusion), HuggingFace Transformers, scikit-learn
- **Cloud & Infrastructure:** GCP (Cloud Run, GKE, Vertex AI), AWS (EC2, Lambda, S3, EKS), Kubernetes, Docker, FastAPI, Spring Boot, CI/CD (Jenkins, SonarQube), MongoDB, RabbitMQ, Kafka, MQTT, Microservices, REST
- **Languages, Frontend & Embedded:** Python, C++, TypeScript, SQL; React, Astro; Linux, Git

PROFESSIONAL EXPERIENCE

Virginia Polytechnic Institute and State University

2024 - 2026

Graduate AI/ML Researcher

Blacksburg, VA

- Architected a LangGraph + Llama 3 multi-agent pipeline for grammar-constrained SDF XML generation, enabling procedural authoring of 100+ simulation environments — stateful task graphs with retry logic and constrained decoding for reliable structured output.
- Designed and experimentally validated the Conflict Architecture (CA) framework for brain-inspired UAV navigation (NSF Award #2204780, PI: J.M. Paul), achieving a simultaneous 34.6% reduction in compute energy and elapsed time over baseline ($p < 0.0001$, $n = 1,000$ runs, 3 real-world-mapped environments).

GE HealthCare

2022 - 2024

Software Engineer II

Bengaluru, IN

- Led the architecture migration of a legacy C++ monolith to Kubernetes-based Java/Spring Boot microservices, cutting application runtime by 80% by designing and implementing image normalization and segmentation algorithms in C++.
- Profiled and optimized a Swin Transformer model end-to-end — ONNX export, FP16 precision tuning, custom CUDA/C++ plugin authoring for unsupported ops, and layer fusion via the TensorRT builder API — for 20% throughput uplift on production hardware, enabling real-time inference.
- Introduced automated quality gates and CI/CD pipelines (Jenkins, SonarQube), reducing release defects by 60% across 12 quarterly production releases.

The Next Move Electronics (TNM)

2020 - 2022

Co-Founder & Sole Technical Lead

Delhi, IN

- Founded and bootstrapped (~\$10K self-funded) a home-automation startup as the sole engineer, owning every layer 0-to-1 — PCB design, firmware, cloud backend, and a Flutter/Alexa mobile app; shipped to 100+ homes before COVID supply-chain disruption ended the venture.
- Co-invented and patented a master-slave microcontroller communication protocol enabling single-chip control of multiple devices — cutting Bill of Materials cost 80% and directly enabling first design-win customers.
- Architected an AWS-backed microservice backend (EC2, Lambda, RabbitMQ, MongoDB, MQTT), scaling 0 to 1000+ devices at 99.9% uptime and cutting field bug reports 98% via a hardware-software validation workflow built solo.

PROJECTS

Agentic Web Automation System

- Built a multi-step agentic automation system using LangGraph orchestration and Browser Use for DOM interaction, powered by a locally-quantized Qwen model on an RTX 5060 Ti — no cloud API dependency.
- Designed stateful task graphs with retry logic, conditional branching, and tool-driven form-filling across multi-page workflows.
- Achieved ~60s end-to-end task completion with sub-300ms first-token latency on local inference.

Professional RAG Pipeline

- Built a two-stage retrieval pipeline (BGE-base embeddings → MS-MARCO Cross-Encoder reranker) evaluated with Hit@K and MRR; LLM-as-judge scoring showed consistent faithfulness gains over single-stage retrieval across 50+ test queries — implemented from scratch without framework abstractions.

- Implemented SHA-256 query hashing for semantic caching, reducing LLM API overhead by 30%; added per-query token/cost tracking for full inference observability.
- Deployed as a containerized microservice on Google Cloud Run with per-stage latency instrumentation and grounded generation to eliminate hallucinations.

EDUCATION

Virginia Polytechnic Institute and State University

Aug 2024 - May 2026

M.S., Computer Engineering (Thesis Track)

Blacksburg, VA

- **Achievements:** Defended M.S. Thesis, [Brain-Inspired Drone Navigation Using the Conflict Architecture](#) — 4.0 GPA
- **Coursework:** CS5424: Advanced Machine Learning, CS5465: Applications of Machine Learning, ECE5504: Computer Architecture

Uttarakhand Technical University

Jul 2016 - Dec 2020

B.E., Electronics and Communication Engineering

Uttarakhand, IN

- **Achievements:** Graduated with First Division

PUBLICATIONS

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- V. Chauhan, [Brain-Inspired Drone Navigation Using the Conflict Architecture](#), M.S. Thesis, Virginia Tech, 2026.
 - D. Dave, G. Sawhney, V. Chauhan, “[Multi-Agent Stock Prediction Systems: Machine Learning Models, Simulations, and Real-Time Trading Strategies](#),” arXiv:2502.15853, 2025.
 - V. Chauhan et al., *Master-Slave Microcontroller Communication System for Home Automation*, The Next Move Electronics — [Patent No. 202111060973](#) (Granted), 2021.

AWARDS & RECOGNITION

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- **GE Impact Award:** Recognized for independently modernizing a critical legacy application to production microservices, improving software delivery at scale.
 - **NSF-Funded Graduate Research Assistant:** Supported on NSF Award #2204780 (PI: JoAnn M. Paul); core contributor to thesis research on brain-inspired autonomous systems.
 - **Honor Society of Phi Kappa Phi:** Inducted, Virginia Tech Chapter (2026); top 10% of Master’s graduate students across all disciplines.
 - **Omicron Delta Kappa National Leadership Honor Society:** Selected for membership, Alpha Omicron Circle, Virginia Tech (2026 Cohort).